

Hamza N. Hassan

Computer Engineering

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Work Experiences:

Software Engineering, Full-Time

Progresssoft

Sep 2024 – present

PayHub :

- Core team member in the **PayHub project** used for banking and payment integrations.
- Implemented **SWIFT MT message structural validation** and financial message parsing according to SWIFT standards.
- Developed **backend services using Java and Spring Boot** for financial transaction processing.
- Built **RESTful APIs** integrated with domino-based frontend applications.
- Implemented **XML and message format validations** (references matching, ISO-8601 date validation, message status verification).
- Designed and implemented **unit and integration tests** using JUnit, Mockito, WireMock, and RestAssured.
- Worked with **PostgreSQL and MySQL databases** for transaction data persistence.
- Contributed to **CI/CD pipelines using GitLab, Docker, and Kubernetes deployments**.
- Applied **Clean Code principles and Agile Scrum practices** in team development.

Payment Messages Agentic Workflow :

- Contributed to a Proof of Concept for an AI-driven coding agent designed to automate financial message processing and replace core functionalities provided by the Provide Core used in banking integrations.
- Designed an **agentic workflow for MT/MX message conversion**, enabling automated translation between SWIFT MT Messages and ISO 20022 MX Messages formats.
- Implemented **message mapping and generation logic** to support bidirectional conversion between MT and MX financial message structures.
- Developed **AI-assisted validation pipelines** capable of validating both MT and MX messages.
- Implemented the entire project using Kotlin, ensuring type safety, scalability, and maintainability of the system.

Software Developer, Full-Time**Leading point****April 2024 – Sep 2024**

- Spearheaded the development of an automation API project using Python and Fast API, significantly reducing manual effort and enhancing workflow efficiency for the process team.
- Streamlined data retrieval processes from Azure DevOps, SharePoint, and SonarQube, enabling team members to concentrate on their core responsibilities.
- Developed “Q-POINT,” a comprehensive Django-based website system for streamlined process management, incorporating frontend technologies such as HTML, CSS, and JavaScript.
- Managed and optimized Postgres databases to support various application needs
- Contributed to the "Vision Point" project by using React to troubleshoot and resolve bugs, as well as designing and implementing new pages to enhance the application's functionality and user experience.

Software Engineering, intern**Wiley / Atypon****May 2023 - Oct 2023****○ Software Engineering:**

- Implemented a decentralized cluster-based NoSQL database using Java, spring boot, and Docker.
- Created a Webapp using different technologies, Sockets, Servlets, and Spring Boot, to demonstrate the difference between each of them and the advancement of web technologies using Java.
- Created an Uno game engine that contains object-oriented programming and a powerful design using Java, design patterns.
- Implemented the old maid card game using Java Multithreading, in which each player is represented as a thread that plays the game.

○ DevOps:

- Proficient in utilizing **Git** and GitHub for version control and collaborative software development.
- Experienced in designing and implementing microservices architecture using **Docker** containers, ensuring scalable and efficient deployment of distributed systems.
- Experienced in **Linux** and shell scripting, creating scripts for detailed file analysis reports in specified directories and subdirectories, enhancing efficiency and data insights.

Education:

Amman, Jordan**University of Jordan****Sep 2019 - Jan 2024**

- Bachelor of Engineering, Computer Engineering
- High school AVG: 95.5

Projects:

decentralized cluster-based NoSQL database :

- Developed a decentralized NoSQL database using Java Spring boot/ RESTful API provide their own APIs for performing queries, instead of using SQL with a focus on load balancing, data consistency, and efficient query processing.
- Implemented a custom bootstrapping node for cluster initialization and user-node assignment.
- Managed inter-node communication via Docker containers where each node has a replica of the DB, avoiding any central points of failure.
- Created a comprehensive demo application for a school registration system, enabling management of teacher and student registrations, grade assignments, class modifications, and more, all within the database.
- Adhered to SOLID principles and Clean Code practices, with thorough testing to validate system performance.
- **Link** <https://github.com/hamzahassancode/Decentralized-Cluster-Based-NoSQL-Database>

Water-pump-Automation :

- Developed a mobile application using Flutter and Dart for monitoring and managing water processes within homes. The app integrates with Firebase and Arduino to monitor tank water levels and solar cells' performance.
- Users can remotely control the water pump and solar cells, with the option to switch between manual and automatic modes.
- Additional features include checking the availability of nearby water tankers and requesting a refill when water levels are low.
- **Link** <https://github.com/hamzahassancode/Flutter-App-Water-pump-Automation>

Cardio-Disease-Prediction-AI:

- predicting cardiovascular diseases by exploring and analyzing a Kaggle dataset with 13 features and 70,000 records.
- Trained and evaluated classical models (RandomForest, SVC, XGBClassifier).
- Developed and fine-tuned a Keras model, achieving 72% accuracy and 88% recall with a threshold of 0.3.
- **Link** <https://github.com/hamzahassancode/Cardio-Disease-Prediction-AI>

Home-Decor-Shop:

- Designed and developed a responsive web application for showcasing a range of home decor.
- Utilized HTML for structuring the content, CSS for styling and creating a visually appealing layout, and JavaScript for interactive elements and dynamic content.
- Implemented a responsive design to ensure seamless navigation and optimal viewing experience across various devices and screen sizes.
- **Link** <https://github.com/hamzahassancode/Home-Decor-Shop-HTML-CSS-JS>

Licenses & certifications:

- **ICPC Jordanian Colligate Programming Contest (2021) | 11th Place Locally - ICPC International Collegiate Programming**
- **The Complete 2022 Flutter & Dart Development Course - Udemey**
- **Certificate of Training course in Software Engineering Using Java and DevOs - Wiley/Atypon**

Technical Skills:

- **Programming Languages:** Dart, Java, C++, Python, Kotlin
- **Web Development:** HTML, CSS, JavaScript, React, Django, Spring Boot, Servlets
- **Database Management:** SQL, NoSQL, MySQL, PostgreSQL, H2 Database (for testing)
- **Tools & Technologies:** Git, Docker, Linux, Kubernetes (MicroK8s), CI/CD, Liquibase
- **Problem Solving:** Strong analytical and problem-solving skills
- **APIs:** RESTful APIs, FastAPI
- **Testing:** Unit Testing with Junit and AssertJ, Integration Testing using RestAssured, MOCKITO